## More to Know

**Photochemistry** is the branch of chemistry which deals with chemical reactions involving light.

### f. Catalyst

Sometimes you are advised by the elders to drink a small amount of 'oma water' after a heavy meal. Do you know, why? This is because oma water makes digestion faster. Likewise, in industries some chemical substances are used to speed up chemical reactions. These substances are called catalysts. For example, metallic iron is used as a catalyst in the manufacture of ammonia using Haber's process. This ammonia is the basic material for the production of urea, an important fertilizer in agriculture. In vanaspati ghee (dalda) preparation, finely divided nickel is used as a catalyst. Thus, speed of certain reactions is influenced by the catalysts and such reactions are called **catalytic reactions**.



**Figure 10.2** Urea applied on paddy crops

DO YOU KNOW?

Enzymes and yeasts are called biocatalysts.



## Activity 3

Buy some fresh yeast from a grocery shop. Prepare a paste of wheat flour with water in a vessel. Add some yeast and leave the vessel closed for few hours under sunlight. Observe the changes closely. What do you infer?

## 10.2 Effects of Chemical changes

We know that every chemical reaction requires a specific condition to occur. When chemical reactions take place there will be production of heat, light, sound, pressure etc. and also many other effects.

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### 10.2.1 Biological Effects

#### a. Spoilage of food and vegetables

Food spoilage may be defined as any change that causes food unfit for human consumption. The chemical reactions catalysed by the enzymes result in the degradation of food quality in the form of development of bad tastes and odour, deterioration and loss of nutrients.

**Examples**

- Rotten eggs develop a bad smell due to formation of hydrogen sulphide gas.
- Decaying of vegetables and fruits due to microbes.

#### b. Rancidity of fishes and meat

Fishes and meat containing high levels of poly unsaturated fatty acids undergo oxidation. It causes bad odour when exposed to air or light. This process is called rancidity.



**Fig 10.3** Rancid fish on the shore



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